

SAMPLE SIZE AND STATISTICAL POWER

Justification for a sample of 300 traders

Developing Gender-Responsive Childcare Services in Uasin Gishu County Markets · Discrete choice experiment

Part 1 — Justification in plain language

This section states the case for a sample of 300 in non-technical terms, for readers who need the reasoning rather than the derivation. Part 2 sets out the formal calculation.

We propose interviewing **300 traders**. Each trader is shown six cards and asked to choose between two childcare services, so 300 traders give us 1,800 separate decisions to learn from — enough to work out what traders genuinely value, without asking any one person to sit through a long interview. We did not settle on 300 by habit. We worked out, for every possible sample size, how sharp the answers would be, and 300 is the point where the study starts producing answers we can stand behind. At 300 traders, if a feature genuinely pulls traders towards one option — enough to tip choices from an even 50-50 split to about 56 out of 100 — we have roughly an **86% chance of detecting it**, rising to **95%** for a slightly stronger preference. Below about 170 traders those chances drop away quickly, and we would risk telling the County that traders showed no clear preference when the truth is simply that the sample was too small to see one. The numbers above 170 buy two further things worth paying for: ordinary findings, such as the share of traders who would actually use the service, come out accurate to within about **6 percentage points** either way; and there are enough women traders — the group this study is really about — to examine on their own rather than only as part of the whole. Allowing for traders who decline and for those who turn out not to care for young children, we would need to approach roughly **360** to finish with 300 completed interviews, which works out at about 50 per market across the six markets.

What 300 traders buys

“Power” is simply the chance that the study spots a real preference rather than missing it.

If a feature tips choices to...	Chance we detect it	Assessment
54 out of 100 (very slight pull)	44%	Unreliable
55 out of 100 (slight)	68%	Still risky
56 out of 100 (modest)	86%	Acceptable
57 out of 100 (clear)	95%	Strong
60 out of 100 (strong)	>99%	Virtually certain

The accepted standard in research is an 80% chance of detection. At 300 traders we clear that standard for any feature that tips choices to about 56 out of 100 or better, which covers the size of preference this study needs to measure.

One sentence for the sceptic. The standard rule of thumb for experiments of this kind asks for at least 167 traders. We are proposing 300, and the extra 133 are not padding — they are what allows women traders to be analysed separately and ordinary percentages to be quoted with confidence.

Two things to flag honestly. These figures assume the corrected choice cards are used; they do not hold for the version used in the pilot. And 300 supports studying one group in depth, not two: if the County wants men

and women compared properly, that needs about 400 with a set quota, and traders with disabilities would need to be recruited deliberately rather than left to chance.

Part 2 — Methodological statement

2.1 Primary approach: design-based calculation

For a multinomial logit model, the asymptotic variance–covariance matrix of the parameter estimates is a known function of the experimental design rather than a quantity requiring simulation (Rose & Bliemer, 2013; de Bekker-Grob et al., 2015):

$$\Omega(\boldsymbol{\beta}, \mathbf{X}) = \left[\sum_{s=1}^S \mathbf{X}'_s (\mathbf{P}_s - \mathbf{p}_s \mathbf{p}'_s) \mathbf{X}_s \right]^{-1}$$

where $\mathbf{X}_s$ is the attribute matrix for choice set s , $\mathbf{p}_s$ the vector of choice probabilities and $\mathbf{P}_s = \text{diag}(\mathbf{p}_s)$. Under null priors ($\boldsymbol{\beta} = 0$) all $J = 3$ alternatives are equiprobable, so $p_j = 1/3$ and the information matrix is exact rather than approximated. The design comprises $S = 18$ choice sets in $B = 3$ blocks, each respondent completing $S/B = 6$ tasks. With N respondents allocated equally across blocks, total information scales as:

$$\Omega_N = \left[\frac{N}{B} \cdot \mathbf{I}_{\text{total}} \right]^{-1} \quad SE(\hat{\beta}_k) = \sqrt{[\Omega_N]_{kk}}$$

Statistical power for $H_0: \beta_k = 0$ against a two-sided alternative at significance level α is:

$$1 - \gamma = \Phi\left(\frac{|\beta_k|}{SE(\hat{\beta}_k)} - z_{1-\alpha/2}\right) + \Phi\left(\frac{-|\beta_k|}{SE(\hat{\beta}_k)} - z_{1-\alpha/2}\right)$$

and the minimum detectable effect at power $1 - \gamma$ is:

$$|\beta_k|_{\min} = (z_{1-\alpha/2} + z_{1-\gamma}) \cdot SE(\hat{\beta}_k)$$

At $N = 300$ the computed standard errors range from 0.060 to 0.101 across the fourteen attribute parameters (median 0.083), with the alternative-specific constant at 0.148. This yields $1 - \gamma = 0.86$ for $\beta = 0.25$ (odds ratio 1.28) and 0.95 for $\beta = 0.30$, with a minimum detectable effect of $\beta = 0.23$ at 80% power and 0.27 at 90%.

Completed interviews	SE range (attributes)	MDE at 80% power	Assessment
150	0.085 – 0.209	0.24 – 0.58	Main effects only
200	0.074 – 0.181	0.21 – 0.51	Workable; one subgroup at a push
300	0.060 – 0.148	0.17 – 0.41	Recommended
400	0.052 – 0.128	0.15 – 0.36	Precision, not new capability

2.2 Secondary approach: established rule of thumb

The Johnson and Orme criterion for choice experiments (Orme, 1998; Johnson & Orme, 2003) requires:

$$N \geq \frac{500c}{t \times a} = \frac{500 \times 4}{6 \times 2} = 167$$

where $c = 4$ is the maximum number of levels on any attribute, $t = 6$ the number of tasks and $a = 2$ the number of non-status-quo alternatives.

2.3 Tertiary approach: G*Power cross-check

G*Power implements no procedure for discrete choice experiments. There is no conditional-logit option in the program, and a DCE power analysis cannot be performed in it directly. The analysis below therefore reframes the question at the level of the individual choice observation, as a one-sample test of a binomial proportion against $\pi_0 = 0.50$ — the marginal representation of a single attribute contrast. This is a legitimate corroborating calculation, and is reported as such rather than as the primary method.

G*Power 3.1 input parameters

Field	Value
Test family	Exact
Statistical test	Proportion: difference from constant (binomial test, one sample)
Type of power analysis	A priori — compute required sample size
Tail(s)	Two
Effect size g	0.06 ($\pi_1 = 0.56$ against $\pi_0 = 0.50$)
α err prob	0.05
Power ($1 - \beta$ err prob)	0.80
Constant proportion	0.50

Using the normal approximation underlying this test:

$$n_{\text{choices}} = \frac{\left[Z_{1-\alpha/2} \sqrt{\pi_0(1-\pi_0)} + Z_{1-\beta} \sqrt{\pi_1(1-\pi_1)} \right]^2}{(\pi_1 - \pi_0)^2} = 543$$

Because the six observations contributed by each respondent are not independent, this figure must be inflated by a design effect, where $m = 6$ is the number of tasks per respondent and ρ the intra-respondent correlation:

$$D = 1 + (m - 1)\rho \quad N = \frac{n_{\text{choices}} \times D}{m}$$

Assumed ρ	Design effect D	Effective choice observations	Required N
0.05	1.25	679	114
0.10	1.50	815	136
0.15	1.75	951	159
0.20	2.00	1,086	181

2.4 Convergence of the three approaches

Approach	Minimum N	Power at N = 300
Design-based (Fisher information)	≈ 170	0.86 at $\beta = 0.25$
Johnson & Orme rule of thumb	167	— (not a power calculation)
G*Power proportion test, $\rho = 0.15$	159	0.97 at $\pi_1 = 0.56$

The three methods converge on a minimum of approximately 160 to 170 respondents; the proposed $N = 300$ exceeds all three. The G*Power route returns *higher* power at $N = 300$ than the design-based calculation, which is expected: the proportion test evaluates a single marginal contrast, whereas the design-based figure accounts for the simultaneous estimation of fifteen parameters from a specific design. **The design-based estimate is the conservative and methodologically appropriate one, and is the figure reported.**

2.5 Precision of descriptive estimates

For prevalence-type outcomes, applying the standard expression for a proportion with $p = 0.5$ at $N = 300$:

$$n = \frac{z_{1-\alpha/2}^2 p(1-p)}{d^2} \Rightarrow d = \pm 5.7\%$$

giving a 95% confidence half-width of ± 5.7 percentage points overall, and ± 7.6 points within a 167-responder subgroup.

2.6 Allocation and attrition

Parameter	Value	Basis
Completed interviews	300	50 per market across six markets
Traders approached	≈ 360	≈ 15% combined refusal and screen-out
Minimum women	200	Keeps the primary population separately modellable
Men	100	Described and compared descriptively, not modelled separately
Purposive disability sample	40 – 60	Recruited through OPDs, additional to the 300
Choice observations	1,800	300 respondents × 6 tasks

2.7 Assumptions and limitations

The design-based calculation assumes null priors, which is conservative — informative priors derived from the pilot would reduce the variance estimates. It further assumes that the corrected experimental design and randomised alternative presentation are implemented; the figures do not apply to the design used in the pilot, in which position and cost were confounded across nine of twelve choice sets. Finally, $N = 300$ supports one stratified subgroup analysis but not two: separate estimation for male traders would require approximately 167 men, implying a total of roughly 400 under a sex quota, and traders with disabilities require purposive over-sampling rather than incidental capture.

References

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Bibliographic details should be verified against the journal of submission before final use. All computed figures in this document were produced in R from the study's own experimental design; the analysis is reproducible from the files supplied with the pilot report.